

Pre NEET Mock Test - 4

(for NEET-2021)

Answer Key

1	(3)	41	(2)	81.	(1)	121.	(2)	161.	(3)
2	(3)	42	(2)	82.	(1)	122.	(4)	162.	(2)
3	(2)	43	(2)	83.	(2)	123.	(4)	163.	(4)
4	(1)	44	(4)	84.	(2)	124.	(3)	164.	(2)
5	(4)	45	(3)	85.	(2)	125.	(2)	165.	(2)
6	(4)	46	(2)	86.	(3)	126.	(1)	166.	(3)
7	(2)	47	(4)	87.	(3)	127.	(1)	167.	(3)
8	(2)	48	(1)	88.	(3)	128.	(3)	168.	(4)
9	(4)	49	(2)	89.	(1)	129.	(4)	169.	(1)
10	(3)	50	(4)	90.	(3)	130.	(2)	170.	(4)
11	(1)	51.	(2)	91.	(3)	131.	(4)	171.	(3)
12	(2)	52.	(1)	92.	(3)	132.	(3)	172.	(1)
13	(3)	53.	(1)	93.	(1)	133.	(1)	173.	(2)
14	(4)	54.	(4)	94.	(1)	134.	(3)	174.	(1)
15	(2)	55.	(1)	95.	(3)	135.	(3)	175.	(4)
16	(1)	56.	(1)	96.	(1)	136.	(3)	176.	(2)
17	(1)	57.	(3)	97.	(2)	137.	(2)	177.	(4)
18	(2)	58.	(2)	98.	(4)	138.	(4)	178.	(3)
19	(1)	59.	(1)	99.	(2)	139.	(2)	179.	(4)
20	(3)	60.	(1)	100.	(3)	140.	(4)	180.	(4)
21	(1)	61.	(3)	101.	(4)	141.	(1)	181.	(1)
22	(3)	62.	(3)	102.	(1)	142.	(3)	182.	(2)
23	(2)	63.	(4)	103.	(4)	143.	(2)	183.	(4)
24	(4)	64.	(4)	104.	(4)	144.	(2)	184.	(3)
25	(3)	65.	(2)	105.	(4)	145.	(3)	185.	(1)
26	(2)	66.	(1)	106.	(3)	146.	(3)	186.	(2)
27	(3)	67.	(4)	107.	(3)	147.	(1)	187.	(4)
28	(2)	68.	(3)	108.	(4)	148.	(3)	188.	(4)
29	(2)	69.	(3)	109.	(3)	149.	(2)	189.	(3)
30	(1)	70.	(4)	110.	(4)	150.	(3)	190.	(2)
31	(4)	71.	(2)	111.	(1)	151.	(1)	191.	(3)
32	(1)	72.	(3)	112.	(2)	152.	(3)	192.	(3)
33	(1)	73.	(3)	113.	(4)	153.	(2)	193.	(2)
34	(1)	74.	(2)	114.	(3)	154.	(4)	194.	(3)
35	(2)	75.	(4)	115.	(1)	155.	(4)	195.	(3)
36	(3)	76.	(4)	116.	(2)	156.	(3)	196.	(1)
37	(2)	77.	(3)	117.	(3)	157.	(3)	197.	(4)
38	(1)	78.	(1)	118.	(4)	158.	(2)	198.	(3)
39	(3)	79.	(2)	119.	(3)	159.	(2)	199.	(4)
40	(2)	80.	(2)	120.	(2)	160.	(2)	200.	(2)

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Hints and Solutions

PHYSICS

SECTION-A

1. Answer (3)

$$\text{Limit of resolution of telescope} = \frac{1.22 \lambda}{D}$$

$$\frac{X}{2000} = \frac{1.22 \lambda}{D}$$

$$X = \frac{1.22 \times 6 \times 10^{-7} \times 2 \times 10^3}{20 \times 10^{-2}}$$

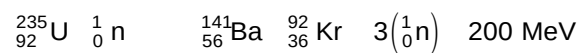
$$X = 6 \times 1.22 \times 10^{-3}$$

$$X = 7.32 \times 10^{-3}$$

X is order of 7 mm

2. Answer (3)

In the given nuclear reaction



From the conservation of nucleons, ${}_{36}^{92}\text{Kr}$ will be emitted

3. Answer (2)

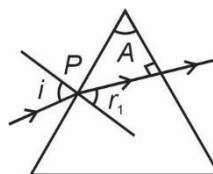
Electric potential due to an electric dipole at any general point is given as

$$V = \frac{kp \cos\theta}{r^2}$$

Where θ is angle with dipole axis.

$$V = \frac{9 \times 10^9 \times 3\sqrt{2} \times 10^{-10}}{9 \times 10^{-2} \times \sqrt{2}} = 30 \text{ V}$$

4. Answer (1)



By Snell's law at point P

$$\frac{\sin i}{\sin r_1} = \mu$$

for small value of θ

$$\sin\theta \approx \theta$$

$$\frac{\sin i}{\sin r_1} = \frac{i}{r_1}$$

$$\frac{i}{r_1} = \mu$$

$$r_1 + r_2 = A$$

$$r_2 = 0$$

$$r_1 = A$$

$$i = \mu A$$

On putting the value of μ and A

$$i = \frac{3}{2} \times 8^\circ$$

$$= 12^\circ$$

5. Answer (4)

Acceleration due to gravity at height h above the surface of earth is given by

$$g' = \frac{gR^2}{(R+h)^2}$$

$$\Rightarrow mg' = \frac{mgR^2}{(R+h)^2}$$

$$mg' = \frac{mgR^2}{(2R)^2}$$

$$mg' = \frac{mg}{4}$$

$$= \frac{80}{4}$$

$$= 20 \text{ N}$$

6. Answer (4)

By Bohr's postulate, angular momentum

$$L = \frac{nh}{2\pi}$$

$$L = \frac{4h}{2\pi} \quad (\because n = 4)$$

$$L = \frac{2h}{\pi}$$

7. Answer (2)

Height of liquid in capillary tube is given as

$$h = \frac{2T \cos \theta}{\rho g r}$$

For water, $\cos \theta = 1$

$$\Rightarrow h \propto \frac{1}{r}$$

$$\text{Now, } \frac{h}{h'} = \frac{r}{2r}$$

$$h' = 2h$$

$$\frac{m'}{m} = \frac{\rho A' h'}{\rho A h} = \frac{A \times 2h}{4Ah} = \frac{1}{2} \quad \left(\because A' = \frac{A}{4} \right)$$

$$m' = \frac{m}{2}$$

8. Answer (2)

The least count of Vernier caliper is given by

$$\text{L.C.} = 1 \text{ M.S.D} - 1 \text{ V.S.D}$$

$$\text{Given, } n \text{ M.S.D.} = 10 \text{ V.S.D}$$

$$\text{So, } 1 \text{ V.S.D} = \frac{n \text{ M.S.D}}{10}$$

$$\text{Hence, } \text{L.C.} = \left(1 - \frac{n}{10} \right) \text{ M.S.D}$$

$$\Rightarrow \frac{1}{10} = \left(1 - \frac{n}{10} \right) 1$$

$$\Rightarrow \frac{n}{10} = \frac{9}{10}$$

$$n = 9$$

9. Answer (4)

We know that

Relative permeability = magnetic susceptibility + 1

$$\mu_r = \chi + 1$$

$$\mu_r = 1 - 0.08 = 0.92$$

10. Answer (3)

The phase angle between displacement and acceleration of a particle executing S.H.M. is π .

therefore $\vec{a} \cdot \vec{r} < 0$

11. Answer (1)

We know that value equivalent to 1 a.m.u in MeV is 931.5

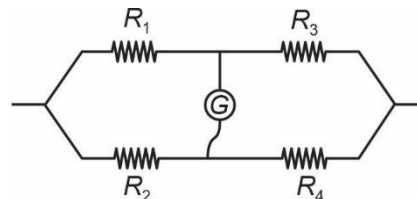
Therefore energy equivalent to 4 a.m.u. is

$$E = 4 \times 931.5 \text{ MeV}$$

$$= 3726 \text{ MeV}$$

12. Answer (2)

In balanced Wheatstone bridge



$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$\text{Here, } R_3 = \frac{50x}{50+x}$$

$$\Rightarrow \frac{50}{\ell_1} = \frac{\left[\frac{50x}{50+x} \right]}{\ell_2}$$

$$\therefore \ell_1 : \ell_2 = 2:1$$

$$\Rightarrow \frac{50}{\ell_1} = \frac{\left[\frac{50x}{(50+x)} \right]}{\left(\frac{\ell_1}{2} \right)}$$

$$2500 + 50x = 100x$$

$$x = 50 \Omega$$

13. Answer (3)

The average thermal energy for gas molecules per degree of freedom = $\frac{1}{2}k_B T$

For diatomic gas, rotational degree of freedom is 2
Average rotational kinetic energy for diatomic gas molecules
= $k_B T$

14. Answer (4)

We know that

$$B_0 = \frac{E_0}{c}$$

$$\frac{B_0}{E_0} = \frac{1}{c}$$

$$\text{As } c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$B_0 = E_0 \sqrt{\mu_0 \epsilon_0}$$

15. Answer (2)

Velocity gradient = $\frac{dv}{dx}$

$$\left[\frac{dv}{dx} \right] = \frac{[M^0 L T^{-1}]}{[L]}$$

$$= [M^0 L^0 T^{-1}]$$

16. Answer (1)

Magnetic field at end of the solenoid is given as

$$B = \frac{\mu_0 n I}{2}$$

$$\Rightarrow B = \frac{4\pi \times 10^{-7} \times 100 \times 5}{2}$$

$$\Rightarrow B = 3.14 \times 10^{-4} \text{ T}$$

17. Answer (1)

When two physical quantities are added or subtracted, there should be as many decimal places retained in the final results as there in the number with the least decimal place. So,

$$23.623 + 8.7 = 32.323$$

$$= 32.3$$

18. Answer (2)

Since frequency is greater than threshold frequency. Hence photoemission will occur.

Now, photoelectric current $\propto$ Intensity of light, so it will be halved

19. Answer (1)

From the colour coding of carbon resistor

Red $\rightarrow$ Digit $\rightarrow$ 2

Yellow $\rightarrow$ Digit $\rightarrow$ 4

Violet $\rightarrow$ Multiplier $\rightarrow 10^7$

Silver $\rightarrow$ tolerance $\rightarrow 10\%$

therefore $R = 24 \times 10^7 \Omega \pm 10\%$

20. Answer (3)

X coordinate of COM is given by

$$X_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3}$$

$$\Rightarrow X_{cm} = \frac{m \times 0 + m \times a + m \times \frac{a}{2}}{3m}$$

$$\Rightarrow X_{cm} = \frac{a}{2}$$

21. Answer (1)

In a transistor base has least concentration of impurity. Emitter is heavily doped.

22. Answer (3)

mean free path of gas molecules is given as

$$\lambda = \frac{1}{\sqrt{2} n \pi d^2}$$

$$\lambda \propto \frac{1}{d^2} \text{ and } \lambda \propto \frac{1}{n}$$

23. Answer (2)

By ideal gas equation

$$PV = nRT$$

$$P = \frac{\rho}{M} RT$$

$$\Rightarrow T = \frac{PM}{\rho R}$$

$$\Rightarrow T = \frac{249 \times 10^3 \times 4 \times 10^{-3}}{0.1 \times \frac{25}{3}}$$

$$\Rightarrow T = 1195 \text{ K}$$

$$\Rightarrow t = 1195 - 273$$

$$= 922^\circ \text{C}$$

24. Answer (4)

Drift velocity (v_d) is given by

$$v_d = \frac{i}{neA}$$

$$\Rightarrow v_d = \frac{8}{5 \times 10^{28} \text{ m}^{-3} \times 1.6 \times 10^{-19} \times 10^{-4}}$$

$$\Rightarrow v_d = 10^{-5} \text{ m/s}$$

25. Answer (3)

By equation of uniformly accelerated motion

$$v^2 = u^2 + 2as$$

$$v = \sqrt{v_0^2 + 2gh}$$

$$\Rightarrow v = \sqrt{(30)^2 + 2 \times 10 \times 80}$$

$$\Rightarrow v = 50 \text{ m/s}$$

26. Answer (2)

Young's modulus of elasticity is

$$Y = \frac{\text{Stress}}{\text{Strain}}$$

$$\text{Strain} \left(\frac{\Delta L}{L} \right) = \frac{Mg}{AY}$$

27. Answer (3)

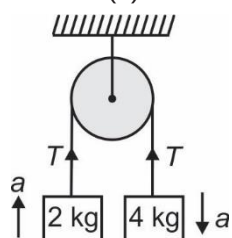
Beats frequency = $f_P \sim f_Q$

$$f_P - 260 = \pm 6$$

$$f_P = 260 \pm 6$$

Thus possible frequency of fork P are $260 \pm 6 = 266 \text{ Hz}$, or 254 Hz . As frequency decreases on loading, therefore original frequency of fork P is 266 Hz .

28. Answer (2)



for 4 kg block

$$4g - T = 4a \quad \dots(i)$$

for 2 kg block

$$T - 2g = 2a \quad \dots(ii)$$

On solving (i) and (ii)

$$T = \frac{2 \times 2 \times 4 \times 10}{6} \Rightarrow T = \frac{80}{3} \text{ N}$$

29. Answer (2)

For an isolated charge capacitor, charge on the capacitor is independent on inserting or removal of dielectric slab, therefore charge on the capacitor remains same

30. Answer (1)

For bright fringe path difference = $n\lambda$

$$(n+1)\lambda_2 = n\lambda_1$$

$$(n+1)420 = n560$$

$$(n+1)3 = 4n$$

$$n = 3$$

31. Answer (4)

Power = Intensity $\times$ area

$$= 10^5 \times 400$$

$$= 4 \times 10^7 \text{ W}$$

32. Answer (1)

The variation of resistivity with temperature is given by

$$\rho = \rho_0 (1 + \alpha t)$$

therefore, on increasing temperature resistivity of conductor increases

33. Answer (1)

Heat required to increase the temperature of a body of mass m by ΔT °C is given by

$$Q = mS\Delta T$$

$$\frac{Q_1}{Q_2} = \frac{S_1}{S_2}$$

$$= \frac{4}{3}$$

34. Answer (1)

Torque of force $\vec{F}$ acting at $\vec{r}$ is given by

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\text{Here } \vec{r} = (0-1)\hat{i} + (0+1)\hat{j} + (0-1)\hat{k}$$

$$= -\hat{i} + \hat{j} - \hat{k}$$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -1 \\ 1 & 2 & 0 \end{vmatrix}$$

$$= [0 - (-2)]\hat{i} - \hat{j}[0 - (-1)] + \hat{k}(-2 - 1)$$

$$= (2\hat{i} - \hat{j} - 3\hat{k}) \text{ N m}$$

35. Answer (2)

From the relation between electric field and potential

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$$

$$\vec{E} = -\frac{\partial V}{\partial y} \hat{j} \left[\text{As } \frac{\partial V}{\partial x} = \frac{\partial V}{\partial z} = 0 \right]$$

$$\Rightarrow \vec{E} = -\frac{\partial}{\partial y} (2y^3) \hat{j}$$

$$\Rightarrow \vec{E} = -6y^2 \hat{j}$$

$$\text{at } y = 1 \text{ m}$$

$$\Rightarrow \vec{E} = -6 \hat{j} \text{ V m}^{-1}$$

SECTION-B

36. Answer (3)

By Brewster's law

$$\mu = \tan i_p$$

$$i_p = \tan^{-1}(\mu)$$

$$= \tan^{-1}\left(\frac{4}{3}\right)$$

$$= 53^\circ$$

37. Answer (2)

In forward bias width of depletion region of p-n junction diode decreases and in reverse bias width of depletion region of p-n junction diode increases. On decreasing the doping concentration depletion width increases.

38. Answer (1)

Electric field at a distance ($r > R$) from the centre of the shell is given by

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$\text{Here, } r = \frac{4R}{3}$$

$$\Rightarrow E = \frac{9q}{64\pi\epsilon_0 R^2}$$

39. Answer (3)

$$\text{As } 1.6 \times 10^{-19} \text{ J} = 1 \text{ eV}$$

$$\therefore 64 \text{ J} = \frac{64}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 4 \times 10^{20} \text{ eV}$$

40. Answer (2)

The capacitive reactance is given by

$$X_C = \frac{1}{\omega C}$$

$$= \frac{1}{100 \times 2 \times 10^{-6}}$$

$$= 5 \times 10^3 \Omega$$

$$\therefore I = \frac{V}{X_C}$$

$$\Rightarrow I = \frac{300}{5 \times 10^3}$$

$$= 60 \times 10^{-3} = 60 \text{ mA}$$

41. Answer (2)

de-Broglie wavelength is given by

$$\lambda = \frac{h}{P}$$

$$\text{As } P = \sqrt{2mE}$$

Hence,

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 320 \times 1.6 \times 10^{-19}}}$$

$$= 0.7 \text{ \AA}$$

42. Answer (2)

In adiabatic expansion heat is neither gained nor lost by the gas and temperature reduces and volume increases.

43. Answer (2)

By de Morgan's law

$$Y = \overline{\overline{A} \cdot \overline{B}}$$

$$= \overline{\overline{A} + \overline{B}}$$

$$= A + B$$

44. Answer (4)

For metals, temperature coefficient of resistance is positive while in insulators and semiconductors temperature coefficient of resistance is negative.

Therefore resistance of insulators and semiconductors decreases on increasing the temperature.

45. Answer (3)

Average power consumed in AC circuit is given by

$$P = I_{r.ms} \cdot V_{r.ms} \cos \phi$$

$$= \frac{I_0}{\sqrt{2}} \frac{V_0}{\sqrt{2}} \cos \frac{\pi}{3}$$

$$= \frac{V_0 I_0}{4}$$

$$= 0.25 V_0 I_0$$

46. Answer (2)

By Lens maker's formula

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = \left(\frac{5}{3} - 1\right) \left(\frac{2}{R}\right) \quad (\text{As lens is equiconvex})$$

$$\Rightarrow \frac{1}{40} = \frac{4}{3R}$$

$$\Rightarrow R = \frac{160}{3} \text{ cm}$$

47. Answer (4)

By Einstein's equation of photoelectric effect

$$h\nu = \phi_0 + KE_{\max}$$

The maximum kinetic energy of emitted electrons in a photoelectric effect is independent of intensity of light.

48. Answer (1)

The position of the particle is given by

$$x = 12t - t^3$$

$$\therefore v = \frac{dx}{dt}$$

$$\therefore v = 12 - 3t^2$$

$$\Rightarrow t^2 = 4 \quad (\text{For } v = 0)$$

$$\Rightarrow t = 2 \text{ s}$$

49. Answer (2)

Limiting static friction is given by

$$f_{\lim} = \mu_s N$$

$$= \mu_s Mg$$

$$= 0.5 \times 4 \times 10$$

$$= 20 \text{ N}$$

$$f_{\lim} > F_{\text{applied}}$$

If applied force is less than limiting friction, then friction force will be equal to applied force.

$$f = F_{\text{applied}}$$

$$f = 12 \text{ N}$$

50. Answer (4)

Moment of inertia of the spherical shell about tangential axis is given by parallel axis theorem as

$$I = I_{\text{cm}} + MR^2$$

$$\therefore I_{\text{cm}} = \frac{2}{3} MR^2$$

$$\therefore I = \frac{2}{3} MR^2 + MR^2$$

$$= \frac{5}{3} MR^2$$

$$\text{As, } I = MK^2$$

$$\Rightarrow MK^2 = \frac{5}{3} MR^2 \Rightarrow K = \sqrt{\frac{5}{3}} R$$

CHEMISTRY

SECTION-A

51. Answer (2)

$$\Delta_r G = \Delta_r G^\circ + RT \ln Q$$

At equilibrium $\Delta G = 0$, $Q = K_{\text{eq}}$.

$$\text{So, } \Delta_r G^\circ = -RT \ln K_{\text{eq}}$$

$$= -8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 300 \text{ K} \times \ln (1.7 \times 10^2)$$

52. Answer (1)

$$\bullet \text{ Number of N atoms in } N_2 = \frac{10}{28} \times 2 \times N_A$$

$$\bullet \text{ Number of O atoms in } O_2 = \frac{10}{32} \times 2 \times N_A$$

$$\bullet \text{ Number of O atoms in } O_3 = \frac{10}{48} \times 3 \times N_A$$

$$\bullet \text{ Number of Cu atoms in Cu(s)} = \frac{10}{63.5} \times N_A$$

53. Answer (1)

- Carbon monoxide is almost water insoluble
- CO has strong tendency to accept oxygen therefore is a strong reducing agent
- Due to lone pair, CO can form complexes
- It hinders the oxygen carrying capacity of blood by forming carboxyhaemoglobin hence poisonous in nature

54. Answer (4)

Electronic configuration of Co $\rightarrow [Ar]3d^7 4s^2$

Electronic configuration of $Co^{2+} \rightarrow [Ar]3d^7$

$\boxed{1\downarrow} \boxed{1\downarrow} \boxed{1\downarrow} \boxed{1\downarrow} \boxed{1\downarrow}$; number of unpaired $e^- = 3$

$$\text{spin only magnetic moment} = \sqrt{n(n+2)} \text{ BM}$$

$$= \sqrt{3(3+2)}$$

$$= \sqrt{15} \text{ BM}$$

$$= 3.87 \text{ BM.}$$

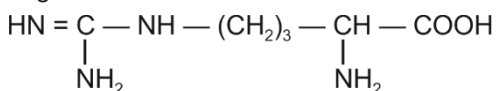
55. Answer (1)

- Cellulose is a natural polymer as its monomer is glucose.
- Neoprene and Buna-S are synthetic rubbers.
- Nylon 6, 6 is a synthetic fibre.

56. Answer (1)

Because of presence of more number of amino groups than carboxyl groups, arginine is basic in nature.

Arginine is



57. Answer (3)

$$n_{\text{H}_2} = \frac{10}{2} = 5 \text{ mol}, n_{\text{O}_2} = \frac{64}{32} = 2 \text{ mol}$$

$$\chi_{\text{H}_2} = \frac{n_{\text{H}_2}}{n_{\text{H}_2} + n_{\text{O}_2}} = \frac{5}{5+2} = \frac{5}{7}$$

$$P_{\text{H}_2} = P \cdot \chi_{\text{H}_2} = 35 \times \frac{5}{7} = 25 \text{ bar}$$

$$P_{\text{O}_2} = (35 - P_{\text{H}_2}) \\ = (35 - 25) = 10 \text{ bar}$$

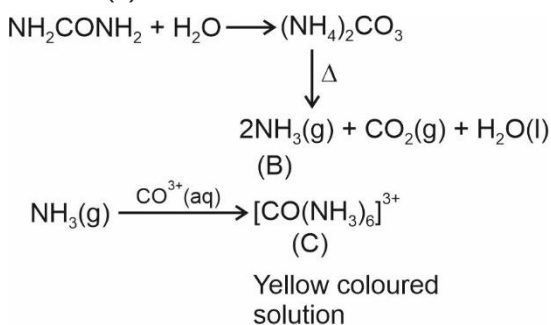
58. Answer (2)

- Thin layer and column chromatography are types of adsorption chromatography
- Paper chromatography is a type of partition chromatography.

59. Answer (1)

- Reaction involving increase in gaseous moles has $\Delta_r S > 0$
- Decomposition reaction is an endothermic process ($\Delta_r H > 0$)

60. Answer (1)



61. Answer (3)

Electrolysis of aqueous NaCl solution using Pt electrodes gives $\text{H}_2(\text{g})$ at cathode and $\text{Cl}_2(\text{g})$ at anode.

62. Answer (3)

- In BCC unit cell, atoms are present at corners and centre of unit cell.
- Shortest interatomic distance in BCC unit cell

$$= \frac{\text{Body diagonal}}{2}$$

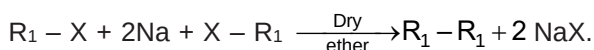
$$= \frac{\sqrt{3}a}{2} = \frac{\sqrt{3}}{2} \times 288 \text{ pm}$$

63. Answer (4)

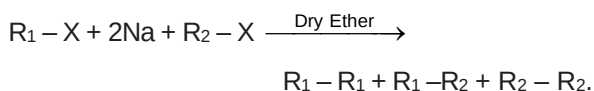
- Inhibitors are substances which reduce the rate of reaction.
- Catalyst provide an alternate pathway or reaction mechanism by reducing the activation energy between the reactants and products.
- Rate constant; $k = Ae^{-E_a/RT}$
So, on increasing temperature rate constant increases.
- Catalyst catalyses the spontaneous reaction but does not catalyse the non-spontaneous reaction.

64. Answer (4)

Wurtz reaction is used to prepare symmetrical alkanes like $\text{R}_1 - \text{R}_1$ as



If R_1 and R_2 are different, then mixture of alkanes may be obtained as



65. Answer (2)

- Oxidation state of Mn in MnO_4^- (permanganate ion) is +7 and in MnO_4^{2-} (Manganate ion) is +6.
- Zn and Cd have completely filled d-subshell therefore not considered as transition metals
- In interstitial compounds, atoms are trapped in interstitial sites therefore non-stoichiometric in nature
- Due to the increasing stability of the lower species to which VO_2^+ and $\text{Cr}_2\text{O}_7^{2-}$ are reduced, oxidising power of VO_2^+ is lesser than of $\text{Cr}_2\text{O}_7^{2-}$

66. Answer (1)

- Cetyltrimethyl ammonium bromide is a cationic detergent.
- Sodium laurylsulphate and sodium dodecyl benzenesulphonate are anionic detergents.
- Stearic acid reacts with polyethylene glycol to form non-ionic detergents.

67. Answer (4)

In benzyl carbocation, the charge is delocalised due to resonance effect hence more stable than secondary butyl carbocation, where charge is delocalised due to hyperconjugation.

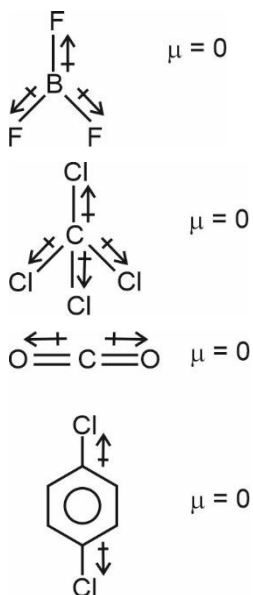
68. Answer (3)

- Ligands having carbon as donor, generally have stronger field strength.
- Order of ligand field strength based on spectrochemical series
 $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$

69. Answer (3)

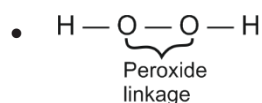
- The iron obtained from blast furnace contains 4% carbon and many impurities in smaller amount. This iron is known as pig iron.
- Cast iron is made by melting pig iron with scrap iron and coke using hot air blast.
- Mond process is used for the refining of Nickel.

70. Answer (4)



71. Answer (2)

- Chlorides and sulphates of magnesium and calcium causes permanent hardness of water.



- $\text{CO} + \text{N}_2$ is producer gas
- CH_4 is an electron precise molecular hydride

72. Answer (3)

Number of protons = atomic number = 58

Number of electrons = Number of protons = 58

Number of neutron = Mass number – Atomic number
 $= 140 - 58 = 82$

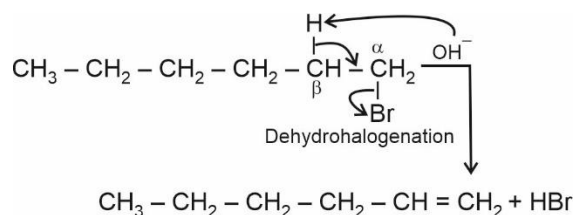
73. Answer (3)

Equivalent wt of aluminium = 9

Number of equivalent of aluminium deposited at cathode = $\frac{9}{9} = 1$

Number of Faraday required = 1

74. Answer (2)

The reaction is also known as β -elimination

75. Answer (4)

- In fullerenes, all the carbon atoms are sp^2 hybridised.
- Silicones are inert therefore used in surgical and cosmetic plants.
- Quartz crystal shows Piezoelectric effect.
- CO_2 is a major contributor in global warming due to greenhouse effect.

76. Answer (4)

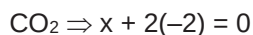
Unnilpentium (105) $\Rightarrow$ DubniumUnnilquadium (104) $\Rightarrow$ RutherfordiumUnnilseptium (107) $\Rightarrow$ BohriumUnunbium (112) $\Rightarrow$ CoperniciumUnnilbium (102) $\Rightarrow$ Nobelium

77. Answer (3)

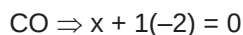
- For non-electrolyte $i = 1$

$$\Delta T_f = iK_f \cdot m = 5.12 \times 0.01 = 0.0512 \approx 0.05 \text{ K}$$

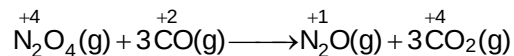
78. Answer (1)



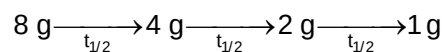
$$x = +4$$



$$x = +2$$



79. Answer (2)

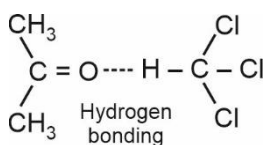


$$\therefore t = 3t_{1/2}$$

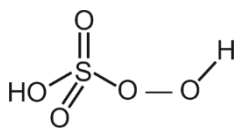
$$t = 3 \times \frac{0.693}{2.303 \times 10^{-2}} = 90 \text{ s}$$

80. Answer (2)

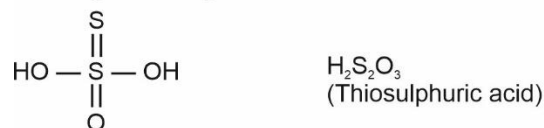
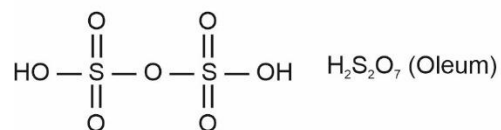
Acetone and chloroform form negative deviation solution as the interaction between acetone-chloroform is stronger than interactions between acetone-acetone and chloroform-chloroform.



81. Answer (1)



Peroxomonosulphuric acid or Caro's acid.

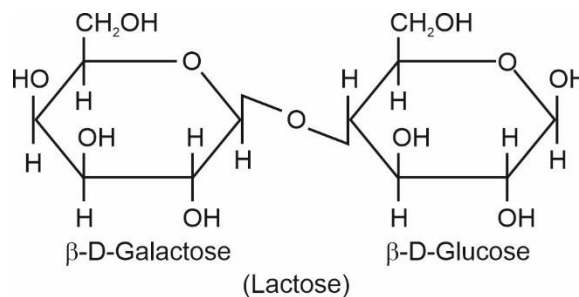


82. Answer (1)

As_2S_3 is negatively charged sol. It can be easily coagulated by cation having higher charge.

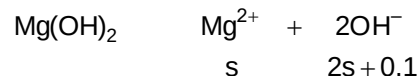
83. Answer (2)

Lactose on hydrolysis gives β -D-glucose and β -D-galactose.



84. Answer (2)

Let the solubility of $\text{Mg}(\text{OH})_2$ be $s \text{ mol L}^{-1}$



[Since $s \ll 0.1$ therefore $2s + 0.1 \approx 0.1$]

$$K_{sp} = [\text{Mg}^{2+}][\text{OH}^-]^2$$

$$1.8 \times 10^{-11} = (s)(0.1)^2$$

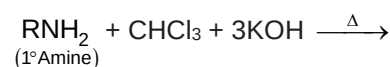
$$s = 1.8 \times 10^{-9} \text{ M}$$

85. Answer (2)

- In free expansion $P_{\text{ext}} = 0$ therefore $w = 0$
- For isothermal process
 $\Delta H = n C_P \Delta T = 0$
 $\Delta U = n C_V \Delta T = 0$
- Entropy increases during the expansion

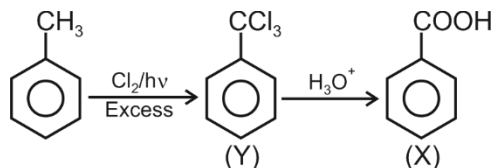
SECTION-B

86. Answer (3)



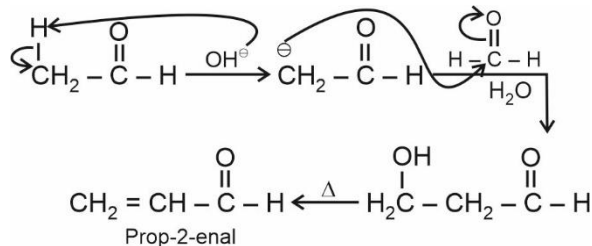
Only 1° amines give carbylamine test.

87. Answer (3)

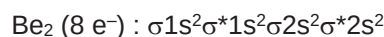


88. Answer (3)

Aldol condensation of two different aldehydes or ketones is known as cross aldol condensation.



89. Answer (1)

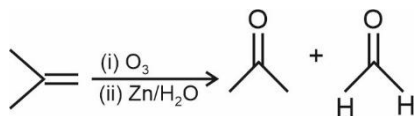


$$\text{BO} = \frac{1}{2} (N_b - N_a) = \frac{1}{2} (4 - 4) = 0$$

$\therefore$ Bond order of $\text{Be}_2 = 0$

$\therefore$ Be_2 doesn't exist.

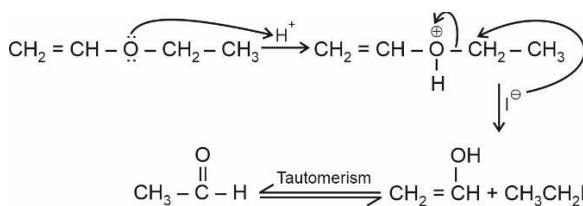
90. Answer (3)



91. Answer (3)

Sodium ions are found primarily on the outside of cells, being located in blood plasma and in the interstitial fluid which surround cells.

92. Answer (3)



93. Answer (1)

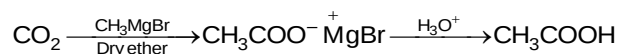
N_2O (Neutral), SO_3 (Acidic)

Na_2O (Basic), PbO (Amphoteric)

94. Answer (1)

Hg cathode is used in Castner Kellner cell as sodium is obtained in the form of sodium amalgam.

95. Answer (3)



96. Answer (1)

In BF_3 , B has incomplete octet therefore it acts as a Lewis acid.

97. Answer (2)

For $l = 0$, possible value of m is 0 only.

98. Answer (4)

• Order of ionic radii $\text{Ce}^{3+} > \text{Pr}^{3+} > \text{Sm}^{3+} > \text{Eu}^{3+}$

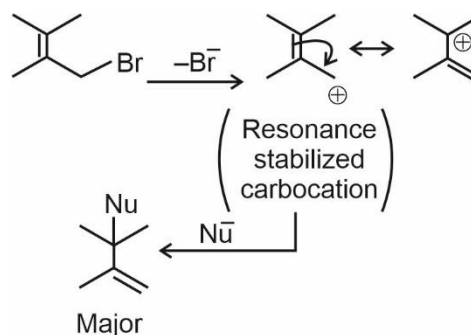
In Lanthanoids, ionic size decreases gradually with increase in atomic number.

99. Answer (2)

NO_2^- is a unidentate ligand having two different donor sites i.e., N and O. Therefore, it is an ambidentate ligand.

100. Answer (3)

• More is the stability of intermediate carbocation faster is the rate of $\text{S}_{\text{N}}1$ reaction.



BOTANY

SECTION-A

101. Answer (4)

Equisetum is a pteridophyte. It has vascular tissues and cones.

As it is homosporous, it does not produce two types of spores.

102. Answer (1)

Thalassemia is quantitative defect in synthesis of globin chains.

Colour blindness is X-linked recessive disorder.

Myotonic dystrophy is autosomal dominant disorder.

103. Answer (4)

Crossing over or recombination takes place in pachytene.

In leptotene, there is condensation of chromatin.

In zygotene, there is synapsis or pairing of homologous chromosomes.

In diplotene, there is dissolution of synaptonemal complex.

104. Answer (4)

Sex of an individual is not an attribute of a population. It is a trait of an individual. Birth rate, death rate, sex ratio are attributes seen in population.

105. Answer (4)

Wood is secondary xylem. It has vessels, tracheids, xylem parenchyma and fibres. Heartwood as well as sapwood both contain vessels and tracheids.

Heartwood is central part of stem. It has tannins, resins, oils etc., provides mechanical support.

Sapwood is peripheral part of wood.

106. Answer (3)

Snow blindness refers to inflammation of cornea due to UV-B radiations.

UV-B also causes various type of skin cancers, aging of skin and cataract of eye.

107. Answer (3)

In the equation $\frac{dN}{dt} = rN \frac{K - N}{K}$

K = carrying capacity

N = population density

r = rate of natural increase

e = base of natural logarithms, (not shown in the equation).

108. Answer (4)

A purine (A, G) pairs with pyrimidine (C, T) with the help of hydrogen bonds.

Guanine is bonded with cytosine with three H-bonds.

Backbone of DNA is constituted by sugar – phosphate.

109. Answer (3)

Those cells which do not divide enter the G_0 phase. In G_0 phase, the cells are metabolically active but do not proliferate.

G_1 , S and G_2 are phases of interphase.

110. Answer (4)

Perigynous flowers have half-inferior ovary. Peach has perigynous flowers.

Mustard, brinjal and potato have superior ovary and hypogynous flowers.

111. Answer (1)

Water lily is an aquatic plant which does not get pollinated by water. It gets pollinated by insects.

Hydrilla, *Zostera* and *Vallisneria* are pollinated by water.

112. Answer (2)

RNA polymerase has ability to transcribe the DNA as well as opening the DNA double helix.

DNA polymerase can replicate the DNA but DNA helicase opens the helix.

Topoisomerase releases tension felt in DNA.

113. Answer (4)

Translation starts when there is binding of mRNA to ribosomes.

Before it, in the first phase there is charging of tRNA.

Formation of peptide bond requires energy.

114. Answer (3)

Replication of DNA takes place in S phase. In G_2 phase there is cell organelle duplication, tubulin protein and RNA synthesis.

115. Answer (1)

Plum, rose and peach have half- inferior ovary.

Guava has inferior ovary and epigynous flower.

Pea has hypogynous flowers.

116. Answer (2)

Agar is obtained from red algae such as *Gelidium* and *Gracilaria*.

Chlorella and *Volvox* are green algae. *Laminaria* and *Sargassum* are brown algae.

Spirulina and *Anabaena* are blue green algae.

117. Answer (3)

Embryo, seed, pollen sac, fruits and ovule are diploid structures.

Pollen grains, female gametophyte and male gametes are haploid structures.

118. Answer (4)

CFCs (chlorofluorocarbons) harm ozone layer as they degrade ozone into O_2 and O atoms. There is good ozone in stratosphere and bad ozone in troposphere.

119. Answer (3)

Length of DNA = No. of base pairs $\times$ distance between two base pairs.

$$1.36 \text{ mm} = x \times 0.34 \text{ nm}$$

$$\frac{1.36 \times 10^6 \text{ nm}}{0.34 \text{ nm}} = x$$

$$4.0 \times 10^6 \text{ bp} = x.$$

120. Answer (2)

Nitrogenase is oxygen sensitive enzyme. It is found in some prokaryotes called N_2 fixer.

It converts N_2 into ammonia. It is Fe-Mo enzyme.

Leghaemoglobin is O_2 scavenger.

121. Answer (2)

Insects are the most species rich taxonomic group.

Among the vertebrates, fishes are most species rich group.

122. Answer (4)

RER has ribosomes for protein synthesis. Lysosome has hydrolytic enzymes such as carbohydrases, lipases, proteases etc.

Golgi modifies proteins but not synthesize them. Peroxisome is a microbody involved in photorespiration.

123. Answer (4)

Pod weight is not the character taken by Mendel for his experiments.

Flower colour, stem height and pod colour were the traits of pea taken by Mendel.

124. Answer (3)

Clostridium butylicum — Butyric acid

Aspergillus niger — Citric acid

Acetobacter aceti — Acetic acid

Lactobacillus — Lactic acid

125. Answer (2)

Prophase — Condensation of chromatin

Metaphase — Chromosomes align themselves at equator

Anaphase — Separation of chromatids

Telophase — Re-appearance of nuclear membrane.

126. Answer (1)

Inclusion bodies of prokaryotes are non membrane bound.

Lysosome, plant vacuole and ER are single membrane bound cell organelles.

127. Answer (1)

The ladybird and dragonflies acts as biocontrol agents. They are useful to get rid of aphids and mosquitoes respectively.

128. Answer (3)

Producers — Grass

Primary consumer — Rabbit

Secondary consumer — Crow

Tertiary consumer — Vulture

129. Answer (4)

Cytokinin helps in cell division and apical dominance.

Ethylene is a gaseous hormone helps in fruit ripening.

GA and ABA are antagonistic to each other.

130. Answer (2)

GAs are used to cause fruits like apple to elongate and improve its shape.

Auxin causes apical dominance and cytokinin counteracts it.

Ethylene is involved in ripening of fruits.

131. Answer (4)

Primary productivity depends on the plant species of an area and variety of environmental factors.

Various species of primary consumers do not decide the primary productivity of plants.

132. Answer (3)

Iron — Required for chlorophyll biosynthesis

Magnesium — Helps to maintain ribosome structure

Chlorine — Maintains anion cation balance

Manganese — Photolysis of H_2O .

133. Answer (1)

Viroids are infectious RNA particles.

They do not have DNA and proteins.

134. Answer (3)

India has 14 biosphere reserves, 90 national parks and 448 wildlife sanctuaries.

135. Answer (3)

Micropyle is opening in the ovule where integuments are absent. It allows pollen tubes to enter in ovule.

Chalaza is the base of ovule. Funicle is the stalk and hilum is the junction of funicle and body of ovule.

SECTION-B

136. Answer (3)

Fibrous roots arise from base of the stems. Primary roots arise from radicle. Lateral roots arise from pericycle.

137. Answer (2)

Substrate level phosphorylation is direct ATP/GTP synthesis.

NADH_2 and FADH_2 are converted to ATP via ETS.

NADPH_2 is reducing power produced during photosynthesis.

138. Answer (4)

Father of Genetics is Mendel.

Sutton and Boveri proposed chromosomal theory of inheritance.

Morgan discovered linkage.

139. Answer (2)

Plasmolysis occurs when a plant cell is placed in a solution that is hypertonic to the protoplasm.

Isotonic solutions have same concentration of solutes.

Hypotonic solutions have less concentration of solutes.

140. Answer (4)

RuBP combines with O_2 to form one molecule of PGA and one molecule of phosphoglycolate and this reaction is catalysed by RuBisCO.

141. Answer (1)

In ABO blood groups the number of genotypes and phenotypes are 6 and 4 respectively.

Genotypes : $\text{I}^{\text{A}}\text{I}^{\text{A}}$, $\text{I}^{\text{A}}\text{i}$, $\text{I}^{\text{B}}\text{i}$, $\text{I}^{\text{A}}\text{I}^{\text{B}}$, ii, $\text{I}^{\text{B}}\text{I}^{\text{B}}$.

Phenotypes : A, B, AB and O.

142. Answer (3)

Primary sludge contains soil and small pebbles. Bacterial flocs are seen in activated sludge.

143. Answer (2)

Laminarin is stored food material in brown algae.

Red algae has floridean starch as stored food material.

Green algae has starch as stored food material.

Blue-green algae have cyanophycean starch.

144. Answer (2)

In monocot leaf there is no differentiation of mesophyll into palisade and spongy. Conjoint vascular bundles are present. Cuticle is present on both sides. Chlorenchyma (mesophyll) is present in the leaves.

145. Answer (3)

PS I is present on stroma lamellae and has reaction center P_{700} .

PS II is present on grana thylakoids along with cyt b₆f and ATP synthase.

146. Answer (3)

Cyclosporin A is used as an immunosuppressive agent in organ transplant patients produced by the fungus *Trichoderma polysporum*. Statins are produced by *Monascus purpureus*. Streptokinase is produced by *Streptococcus*.

147. Answer (1)

The pioneer community in hydrarch succession is phytoplanktons.

Lichens and mosses are pioneer community in xerarch succession. Ferns are pteridophytes.

148. Answer (3)

Egg cell and synergids both are haploid but egg cell is one in number whereas synergids are two in egg apparatus.

149. Answer (2)

Trimerous flowers and tepals are seen in family Liliaceae.

Persistent calyx and epipetalous conditions are seen in Solanaceae.

Inferior ovary is seen in ray florets of sunflower.

150. Answer (3)

Ribosomes are smallest cell organelle not bound by any membrane.

They are found in mitochondria & chloroplasts. They are also called Palade particles, discovered by George Palade.

ZOOLOGY

SECTION-A

151. Answer (1)

- End of T wave marks the end of ventricular systole.
- P wave represents depolarization of auricles.
- T wave represents repolarisation of ventricles.

152. Answer (3)

- Secondary oocyte forms zona pellucida around it.
- Bulbourethral glands or Cowper's glands are paired male accessory glands.
- Interstitial cells or Leydig cells synthesise and secrete androgens.

153. Answer (2)

GIFT, AI and IUI involves *in-vivo* fertilization.

154. Answer (4)

High concentration of LH will cause release of ovum from Graafian follicle (ovulation).

155. Answer (4)

Mucosa is the innermost layer of GIT.

Ileum is a part of small intestine while ilium is a part of coxal bone.

Vermiform appendix is a tubular projection from caecum.

156. Answer (3)

Example of unicellular gland is goblet cells of alimentary canal.

157. Answer (3)

S.L. Miller produced amino acids by mixing CH₄, H₂, NH₃ and water vapour at 800°C.

158. Answer (2)

The infective stage of *Plasmodium* for humans is sporozoite.

159. Answer (2)

Ligases join the two DNA molecules.

160. Answer (2)

Three ganglia in the thorax and six in the abdomen.

161. Answer (3)

ori site is responsible for controlling the copy number of the linked DNA.

162. Answer (2)

Bt toxin is encoded by *cry* gene.

Taq polymerase, a heat stable enzyme is isolated from bacterium *Thermus aquaticus*. This DNA polymerase is used in PCR.

Agrobacterium tumefaciens is considered as natural genetic engineer for plants. Cohen and Boyer accomplished the construction of first recombinant DNA by using native plasmid of *S. typhimurium*.

163. Answer (4)

Elephantiasis or filariasis is a helminthic disease.

Malaria is caused by *Plasmodium*, a protozoan while enteric fever or typhoid is caused by *S. typhi*.

164. Answer (2)

The secretions of the acrosome help the sperm enter into the cytoplasm of the ovum through zona pellucida and the plasma membrane that induces the completion of meiotic division.

165. Answer (2)

DNA stained with EtBr under UV light produces bright orange color bands.

166. Answer (3)

Platyhelminthes *e.g.*, *Fasciola* is a triploblastic acoelomate.

Ascaris is a triploblastic pseudocoelomate.

Pheretima is a triploblastic coelomate.

Aurelia is an acoelomate.

167. Answer (3)

Peptide bonds are present in proteinaceous biomolecules. Nucleic acid has phosphodiester bonds.

Polysaccharides such as chitin and cellulose have glycosidic bonds between their monomeric units.

168. Answer (4)

During inspiration, there is increase in volume of thoracic chamber.

169. Answer (1)

B and T lymphocytes are associated with immune responses of the body.

Basophils secrete serotonin, heparin and histamine. Neutrophils are phagocytic and most abundant WBCs while eosinophils resist infections.

170. Answer (4)

Vertebrates hearts or brains exhibit homology / divergent evolution.

171. Answer (3)

Simple squamous epithelium lines walls of blood vessels, air sacs of lungs etc.

172. Answer (1)

Placoid scales are present in cartilaginous fishes such as Saw fish (*Pristis*).

Exocoetus is a bony fish while *Petromyzon* is a cyclostome.

173. Answer (2)

RuBisCO is the most abundant protein in whole of the biosphere.

Cellulose is the most abundant carbohydrate in biosphere while collagen is the most abundant protein in animal world.

174. Answer (1)

Palindrome in DNA is a sequence of base pairs that reads same on the two strands when orientation of reading is kept the same.

175. Answer (4)

Australian marsupials exhibit adaptive radiation.

176. Answer (2)

Collar bone is clavicle that articulates with acromion process.

Scapula is a large triangular flat bone that has a slightly elevated ridge called spine which projects as flat, expanded process called acromion. Last two pairs (11th and 12th) of ribs are not connected ventrally to sternum and are called floating ribs.

177. Answer (4)

Trichomoniasis and malaria are caused by protozoans.

178. Answer (3)

RE can produce blunt or sticky ends and cleaves DNA by inspecting the length of a DNA sequence.

179. Answer (4)

Low pCO₂, H⁺ ion concentration and temperature and high pO₂ favours the formation of oxyhaemoglobin.

180. Answer (4)

ANF opposes RAAS. ADH prevents diuresis.

181. Answer (1)

Specific Bt toxin genes were isolated from *B. thuringiensis* and incorporated into several crop plants. The choice of genes depends on the crop and the targeted pest. Most Bt toxins are insect-group specific.

182. Answer (2)

A new breed of sheep '*Hisardale*' was developed using Bikaneri ewes and Marino rams by cross breeding.

183. Answer (4)

CNS is dorsal, hollow and single in chordates. In vertebrates, notochord converts into bony or cartilaginous vertebral column.

184. Answer (3)

Hormones from adrenal medulla are called emergency hormones such as adrenaline and noradrenaline.

185. Answer (1)

Pleurobrachia is a ctenophore. *Buthus* is an arthropod.

Ophiura is an echinoderm.

SECTION-B

186. Answer (2)

Pepsin and rennin are gastric proteolytic enzymes. Enterokinase of succus entericus converts inactive trypsinogen into active trypsin.

187. Answer (4)

RNAi is a method that involves silencing of a specific mRNA due to complementary dsRNA molecule that binds to and prevent translation of the mRNA (Silencing).

188. Answer (4)

Valine is a neutral amino acid.

189. Answer (3)

Presence of glucose and ketone bodies in urine are indicative of diabetes mellitus.

190. Answer (2)

Vestibular apparatus is located above cochlea, a coiled portion of labyrinth.

191. Answer (3)

Glucose, valine and chlorophyll are primary metabolites.

192. Answer (3)

Mature insulin lacks C peptide chain and has one intra and two interchain disulfide bridges between A and B chains.

193. Answer (2)

IgG imparts passive immunity to foetus while IgA provides passive immunity to infant.

194. Answer (3)

Haeckel proposed "Ontogeny recapitulates phylogeny" i.e. embryological evidences of evolution which is disapproved by Karl Ernst von Baer.

195. Answer (3)

Abrin is a toxin, a secondary metabolite. Malonate and oxaloacetate are competitive inhibitor of succinic dehydrogenase.

Exoskeleton of arthropods is made up of chitin which is a homopolymer of NAG. GLUT-4 is a proteinaceous receptor molecules for the transport of glucose into cells.

196. Answer (1)

Diabetes mellitus is caused due to insulin deficiency. Insulin is a pancreatic hormone.

Gigantism is due to over secretion of GH before puberty.

Exophthalmic goitre is due to hyperthyroidism while aldosteronism is due to excessive secretion of aldosterone from adrenal cortex.

197. Answer (4)

Midbrain: Cerebral aqueduct, corpora quadrigemina

Hindbrain: Cerebellum

198. Answer (3)

Homo erectus arose about 1.5 mya.

Neanderthal man lived between 1,00,000 - 40,000 years back.

199. Answer (4)

Lippes loop is an inert IUD.

CuT and multiload 375 are copper releasing IUDs.

Progestasert and LNG-20 are hormonal IUDs.

Saheli is a non-steroidal oral contraceptive pill.

200. Answer (2)

Dolly was the first cloned sheep while ANDi was the first genetically modified monkey. Tracy was the first transgenic sheep to produce α -1-antitrypsin.

